SEMINAR III · WEEK 3 · FOUNDATIONS

Numbers & Operations

Learn → Practice → Check → Analyze → Retry → Apply

Shortened week (Labor Day, Mon Sept 7) — four instructional days.

DAY 1

Positive, Negative & Absolute Value

What does the sign mean? Which number is greater? How far from zero?

NUMBERS TELL A STORY

The sign communicates meaning.

+$50 gained

−$50 owed

8°F above zero

−8°F below zero

+100 ft elevation

−100 ft below sea level

Have students interpret the sign in context **before** calculating anything.

WHICH NUMBER IS GREATER?

Farther right on the number line → greater.

$-8 < -2 < \boxed{0} < 3 < 10$

Every positive number is greater than every negative number. Zero is neither positive nor negative.

COMMON MISCONCEPTION

Is −12 greater than −5?

> *No — on the number line, −12 is farther left.*

Debt framing: would you rather owe $5, or owe $12? A balance of −$5 is greater than −$12.

ABSOLUTE VALUE

Distance from zero.

$|-8| = 8$ $|8| = 8$

−8 and 8 are different numbers — but both are eight units from zero. Value and distance-from-zero are not the same idea.

COMMON MISCONCEPTION

"Absolute value just means make it positive."

> The temperature is −4°F. $|-4| = 4$. This does **not** mean the temperature becomes 4°F.

It means −4°F is four degrees away from zero — the shortcut can hide the real meaning.

EXIT TICKET — DAY 1

1. Compare two negative numbers.
2. Find an absolute value.
3. Interpret a negative value in context.

DAY 2

Operations & Order

Adding and subtracting with negatives · choosing the right operation · order of operations

ADDITION WITH NEGATIVES

Think of it as movement on the number line.

- **−5 + (−3):** start at −5, move 3 more negative → −8
- **−5 + 8:** start at −5, move 8 positive → 3

SUBTRACTION WITH NEGATIVES

$$8 - (-3) = 11$$

Subtracting a negative removes a negative quantity. **Context:** removing a $3 loss improves your position by $3.

TEACHER THINK-ALOUD

"Watch me work through this one."

> A bank account has $45. A $62 charge is processed. What's the new balance?

STOP $45 balance, $62 charge · **FIND** new balance · **CONNECT** a charge decreases balance · **TRY** 45 − 62 = −17 · **CHECK** a $62 charge on $45 should go below zero — it does.

Answer: −$17

MULTIPLYING & DIVIDING NEGATIVES

Same Signs → Positive

Different signs → negative. Applies to multiplication *and* division.

> Doesn't transfer to addition/subtraction — that's a different rule.

WHICH OPERATION DO I NEED?

Keywords are clues, not rules.

1. What quantities do I have?
2. What am I trying to find?
3. How are the quantities related?

"Total means add" and "each means multiply" are helpful clues — but the situation decides, not the word.

WHAT HAPPENS FIRST?

> $6 + 3 \times 4$

- **Wrong:** $6+3=9$, then $9\times4=36$
- **Right:** $3\times4=12$, then $6+12=$**18**

COMMON MISCONCEPTION

"Multiplication always comes before division."

$$20 \div 5 \times 2$$

Multiplication and division have **equal priority** — work left to right: $20 \div 5 = 4$, then $4 \times 2 =$ **8** (not 2).

SAME IDEA, ADDITION & SUBTRACTION

$12 - 5 + 3$ = left to right = **10**

PEMDAS doesn't mean "all addition before all subtraction."

EXIT TICKET — DAY 2

1. One integer operation
2. One order-of-operations problem
3. One error-analysis question

DAY 3

Estimation, Calculators & Reasonableness

About what should the answer be — before you trust the exact one?

ESTIMATE BEFORE YOU TRUST

$49 \times 21 \;\rightarrow\;$ round to $50 \times 20 \;\rightarrow\; \approx 1{,}000$

If a calculator shows 10,290, that should immediately look wrong.

ACTIVITY — WHICH ANSWER COULD BE RIGHT?

$49 \times 21 = ?$

A. 102.9   B. 1,029   C. 10,290   D. 102,900

B — 1,029

$50 \times 20 \approx 1{,}000$, so only B is reasonable. Ask: which choices could we eliminate without calculating exactly?

A CALCULATOR IS NOT A CORRECTNESS CHECK

Intended: 52 × 18. Entered: 52 × 81.

The calculator correctly answers the **wrong** calculation. Estimate first, then check what was actually entered.

THE WEEK 3 CHECKLIST

Operation → Size → Sign → Unit → Context

This becomes a standing reference — use it before accepting any answer as final, not just this week.

EXIT TICKET — DAY 3

Why can't an incorrect calculator result ever be "reasonable"?

DAY 4

Apply, Analyze, Retry

Mixed practice, error analysis on missed items, retry after feedback — independent work, not new instruction.

- Mixed retrieval: comparison, operations, order of operations, estimation, reasonableness
- Students classify missed questions — don't default to "execution error" for everything
- Retry after reviewing an example or getting feedback
- Week 3 check, then reflection: one strength, one developing skill, one error, what you did about it

TRANSITION TO WEEK 4

Fractions & Decimals

Same cycle: Learn → Practice → Check → Analyze → Retry → Apply. Keep retrieving operation choice, order of operations, estimation, and reasonableness.